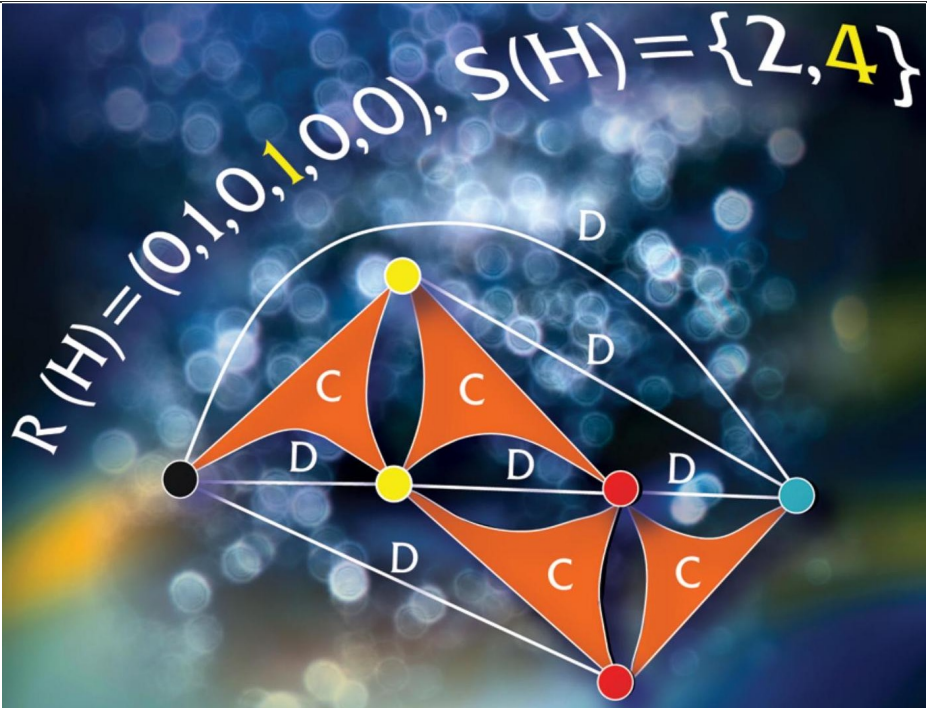
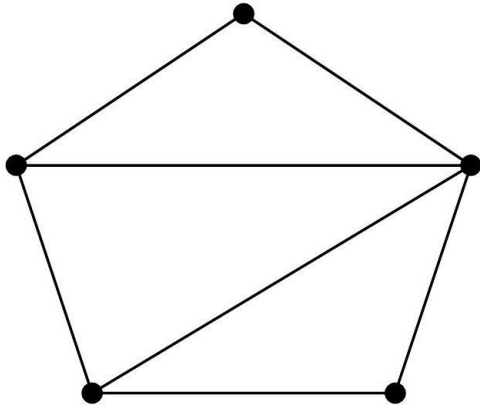
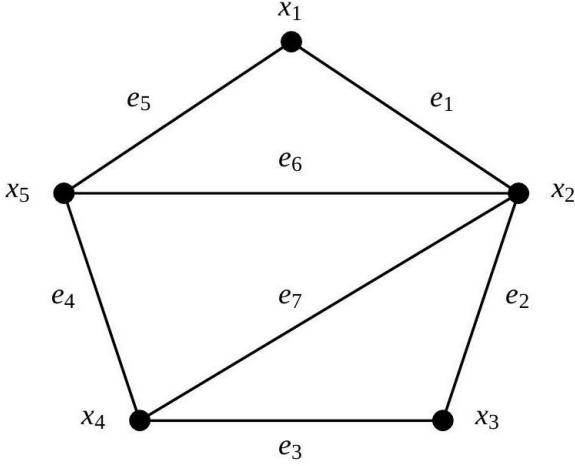
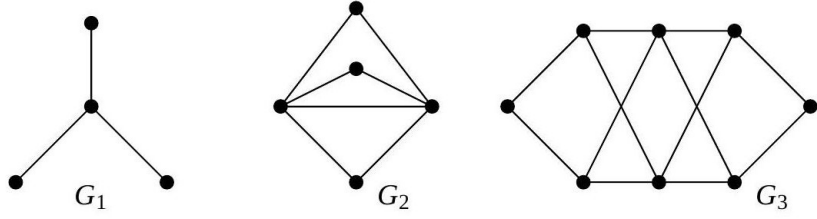
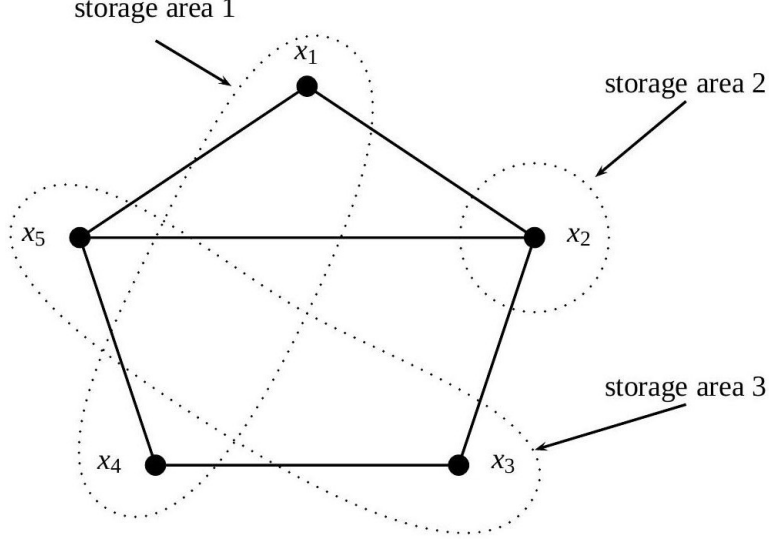
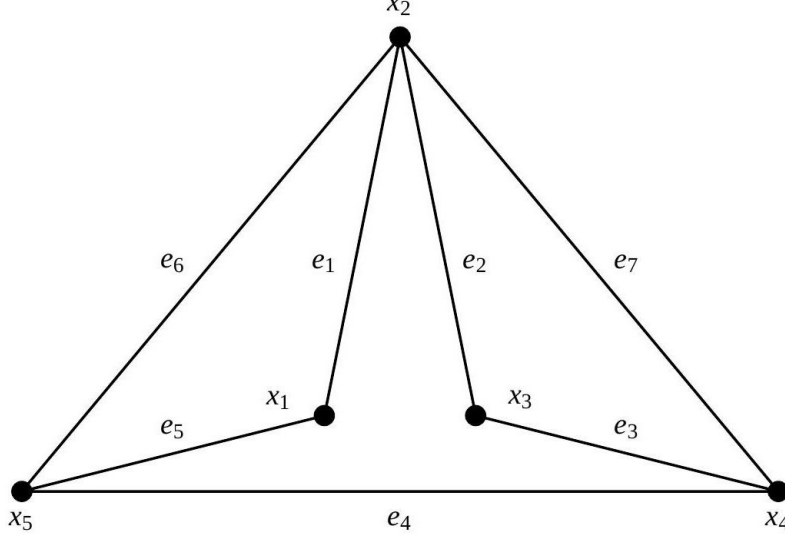
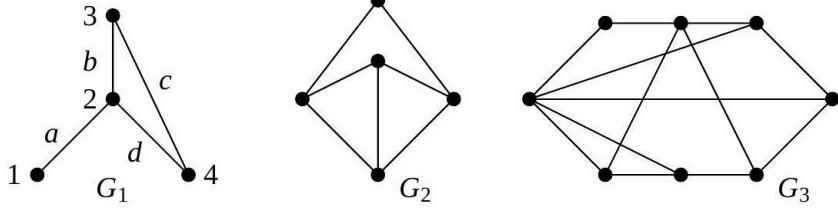
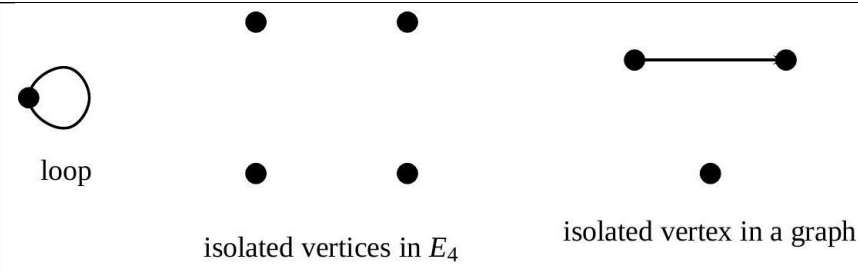
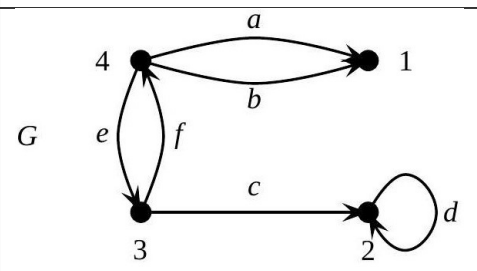
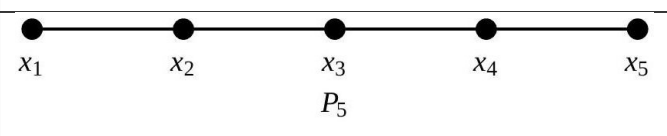
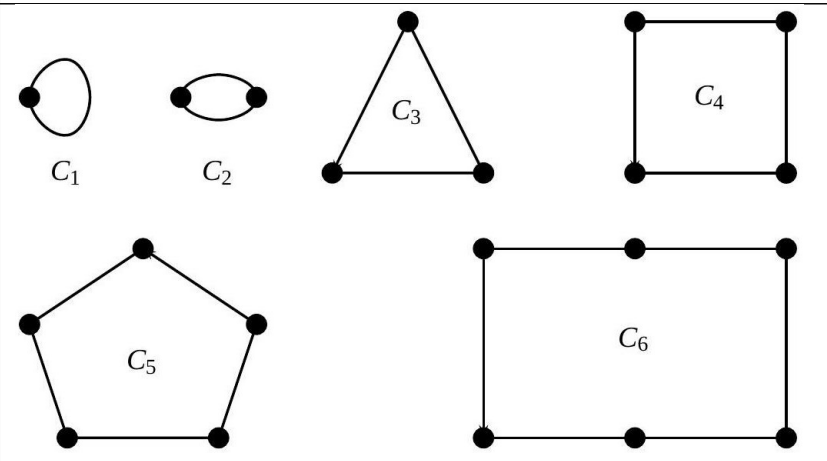
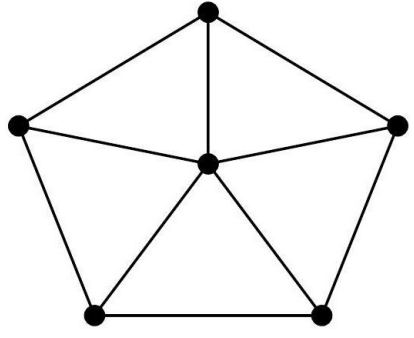
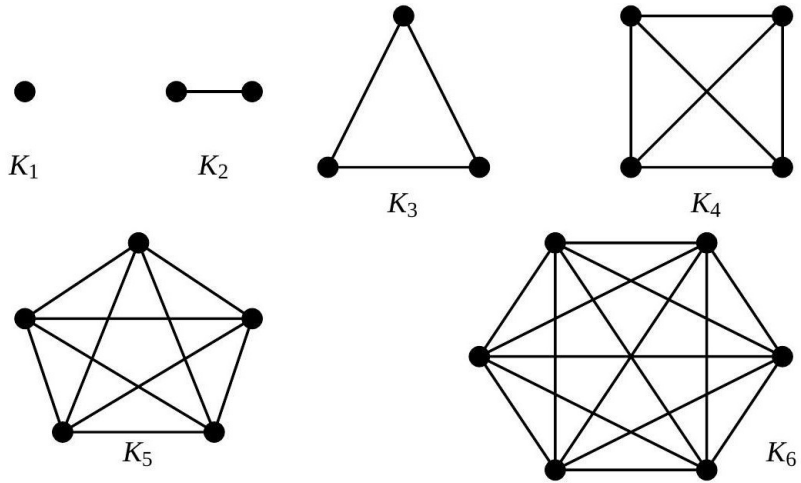
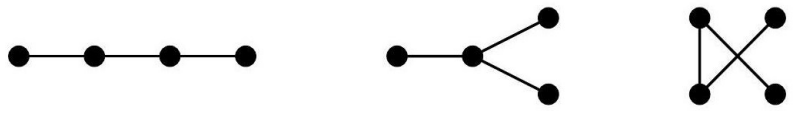
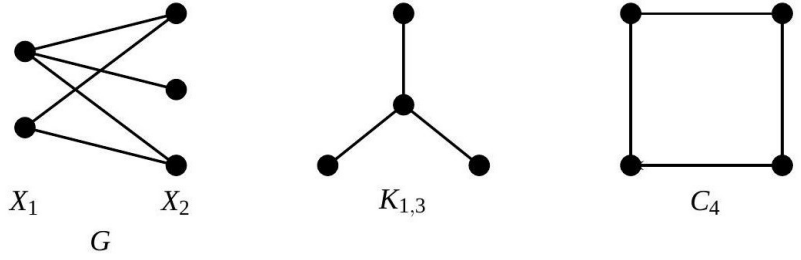
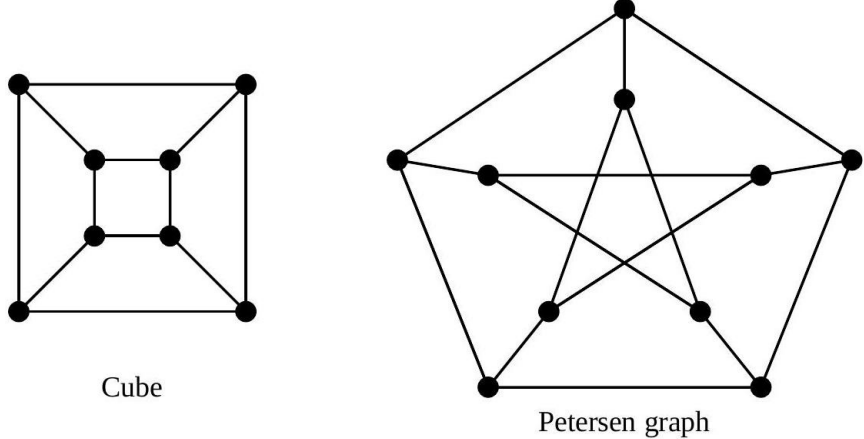


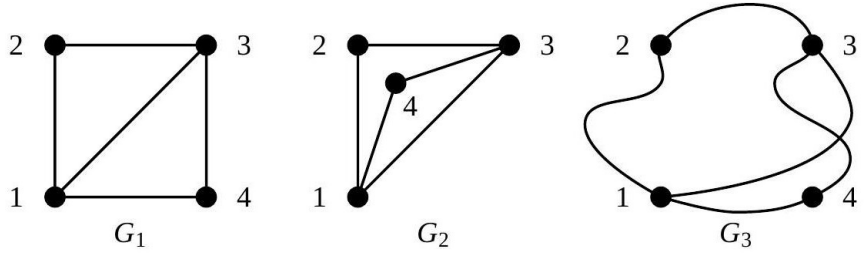
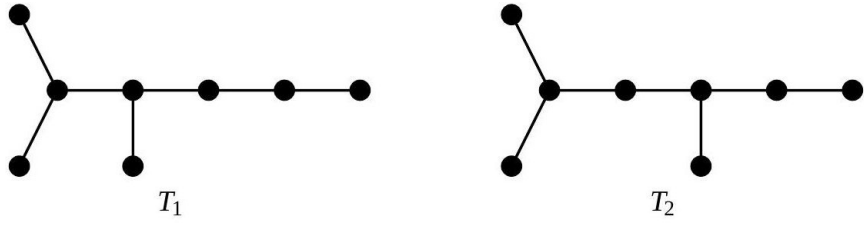
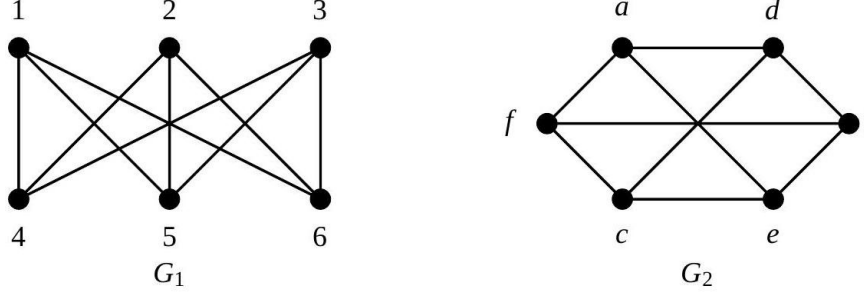
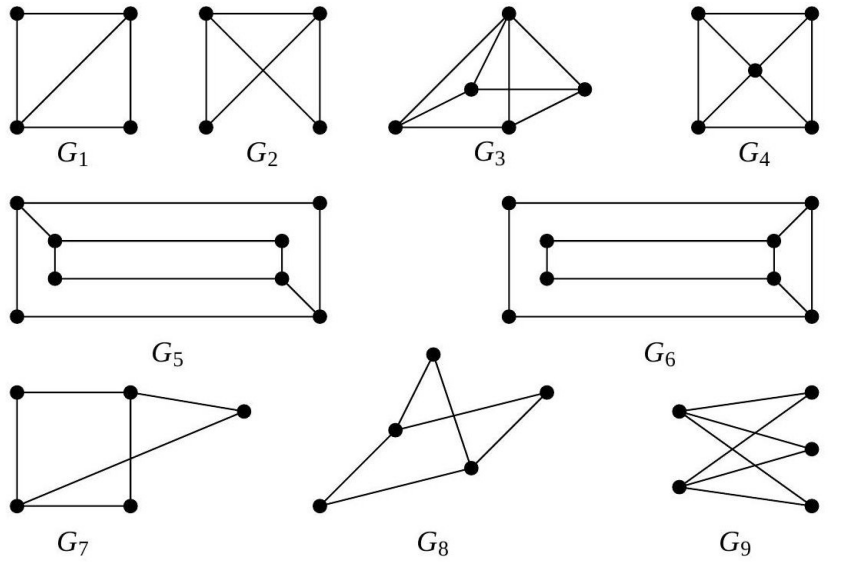
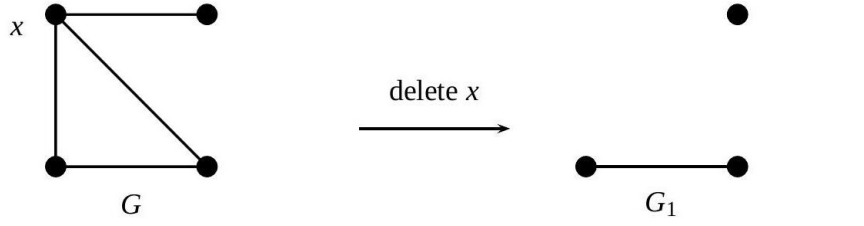
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0009	32	—	—	—	
voloshin-hypergraph_ DIA_0010	33	—	—	—	 G
voloshin-hypergraph_ DIA_0011	34	—	—	—	 P_5
voloshin-hypergraph_ DIA_0012	34	—	—	—	 G_1 G_2
voloshin-hypergraph_ DIA_0013	35	—	—	—	 C_1 C_2 C_3 C_4 C_5 C_6

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0016	36	—	—	—	
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voloshin-hypergraph_ DIA_0018	38	—	—	—	

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voloshin-hypergraph_ DIA_0033	51	—	—	—	

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voloshin-hypergraph_ DIA_0038	56	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0042	61	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0043	64	—	—	—	Diagram illustrating a hypergraph structure. Two sets, X and Y, are shown. X contains S and $X - S_0$. Y contains $N_G(S)$ and $N_G(S_0)$. The region between S_0 and $N_G(S_0)$ is labeled G_A. The region between S and $N_G(S)$ is labeled G_B. Lines connect S to $N_G(S_0)$ and S_0 to $N_G(S)$.
voloshin-hypergraph_ DIA_0044	65	—	—	—	Diagram illustrating a hypergraph structure. Two sets, X and Y, are shown. X contains S and A. Y contains $N_{G_1}(S)$ and B. The region between S and $N_{G_1}(S)$ is labeled G_1. The region between A and B is labeled G_2. A dashed line separates G_1 and G_2.
voloshin-hypergraph_ DIA_0045	66	—	—	—	Diagram illustrating a grid of points connected by lines, forming a rectangular structure. The grid consists of 4 rows and 6 columns of points. Lines connect adjacent points horizontally and vertically, forming a series of rectangles.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0046	68	—	—	—	
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voloshin-hypergraph_ DIA_0049	72	—	—	—	
voloshin-hypergraph_ DIA_0050	74	—	—	—	
voloshin-hypergraph_ DIA_0051	75	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0059	81	—	—	—	
voloshin-hypergraph_ DIA_0060	81	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0062	84	—	—	—	
voloshin-hypergraph_ DIA_0063	86	—	—	—	
voloshin-hypergraph_ DIA_0064	87	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0065	88	—	—	—	The diagram illustrates the construction of a hypergraph from two base graphs, K_5 and $K_{3,3}$. Top section (K_5): A complete graph K_5 with vertices 1, 2, 3, 4, 5 is shown. It is transformed into a triangle with vertices 1, 2, 3. Vertex 4 is placed either 'inside' or 'outside' the triangle. In both cases, the resulting graph has 'no face for 5'. Bottom section ($K_{3,3}$): A complete bipartite graph $K_{3,3}$ with vertices 1, 2, 3, 4, 5, 6 is shown. It is transformed into a square with vertices 1, 2, 3, 4. Vertex 5 is placed either 'inside' or 'outside' the square. In both cases, the resulting graph has 'no face for 6'.
voloshin-hypergraph_ DIA_0066	91	—	—	—	Two graphs are shown side-by-side: G_1: A graph with 10 vertices and many edges, including a cycle of length 6. G_2: A graph with 10 vertices and 12 edges, including a cycle of length 6.
voloshin-hypergraph_ DIA_0067	92	—	—	—	A graph with 10 vertices and 12 edges, including a cycle of length 6.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0068	93	—	—	—	
voloshin-hypergraph_ DIA_0069	97	—	—	—	
voloshin-hypergraph_ DIA_0070	99	—	—	—	
voloshin-hypergraph_ DIA_0071	100	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0072	101	—	—	—	Diagram illustrating the decomposition of a hypergraph G into two hypergraphs G_1 and G_2. G consists of two vertices x and y connected to a set of three internal vertices. G_1 is identical to G but with a dashed edge between x and y. G_2 is a hypergraph where the two vertices x and y are merged into a single vertex xy, which is connected to the same three internal vertices. The equation $G = G_1 + G_2$ is shown.
voloshin-hypergraph_ DIA_0073	102	—	—	—	Diagram illustrating the decomposition of a tree G into two components G_1 and G_2. G is a tree with three vertices x, y, and z. G_1 is a triangle K_3 with vertices x, y, and z. G_2 is a single edge K_2 between vertices x and z. The equation $G = G_1 + G_2$ is shown.
voloshin-hypergraph_ DIA_0074	103	—	—	—	Diagram illustrating a hierarchical tree structure G. The root node G branches into two nodes G_1 and G_2. G_1 branches into G_3 and G_4, while G_2 branches into G_5 and G_6. Each of these nodes further branches into two leaf nodes. Below the tree, a sequence of components $K_{i_1}, K_{i_2}, \dots, K_{i_s}$ is shown, representing the final decomposition of the structure.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0075	104	—	—	—	
voloshin-hypergraph_ DIA_0076	105	—	—	—	
voloshin-hypergraph_ DIA_0077	107	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0079	109	—	—	—	
voloshin-hypergraph_ DIA_0080	111	—	—	—	
voloshin-hypergraph_ DIA_0081	112	—	—	—	
voloshin-hypergraph_ DIA_0082	113	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0084	117	—	—	—	
voloshin-hypergraph_ DIA_0085	119	—	—	—	
voloshin-hypergraph_ DIA_0086	120	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0087	121	—	—	—	
voloshin-hypergraph_ DIA_0088	122	—	—	—	
voloshin-hypergraph_ DIA_0089	124	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0090	125	—	—	—	The diagram shows a transformation from a single edge (two vertices connected by a line) to a more complex graph. The resulting graph has five vertices and several edges, with one cycle highlighted by a dotted line.
voloshin-hypergraph_ DIA_0091	125	—	—	—	A complex graph with 10 vertices and many edges, forming a dense, interconnected structure.
voloshin-hypergraph_ DIA_0092	126	—	—	—	Two graphs are shown side-by-side. The left graph, labeled G, has five vertices labeled 1, 2, 3, 2, 3 and is enclosed in a dotted oval. The right graph, labeled $\bar{G}$, has five vertices and is also enclosed in a dotted oval.
voloshin-hypergraph_ DIA_0093	129	—	—	—	Two transformations are shown. The top transformation shows a graph G with vertices a, b, c, d, e and edges a, b, c, d, e being transformed into a graph $L(G)$ with vertices a, b, c, d, e and edges a, b, c, d, e. The bottom transformation shows a graph with vertices 1, 2, 3, 4 and edges 1, 2, 3, 4 being transformed into a graph with vertices 1, 2, 3, 4 and edges 1, 2, 3, 4. The labels 'edge coloring' and 'vertex coloring' are present below the respective graphs.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0094	130	—	—	—	
voloshin-hypergraph_ DIA_0095	131	—	—	—	
voloshin-hypergraph_ DIA_0096	133	—	—	—	
voloshin-hypergraph_ DIA_0097	134	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0098	136	—	—	—	
voloshin-hypergraph_ DIA_0099	140	—	—	—	
voloshin-hypergraph_ DIA_0100	141	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0101	145	—	—	—	The image displays three network flow diagrams arranged vertically, connected by downward arrows, illustrating the progression of flow from an initial state F_0 to a final state F_2. Diagram 1 (Initial flow F_0): The network consists of nodes u (source), x_1, x_2, x_3, x_4, and v (sink). The edges and their flow/capacity pairs are: $u \rightarrow x_1$ (0,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (1,7), $x_1 \rightarrow x_3$ (1,1), $x_2 \rightarrow v$ (1,6), $x_3 \rightarrow v$ (0,2), and $x_4 \rightarrow x_3$ (1,4). Diagram 2 (Flow F_1): The flow values have updated: $u \rightarrow x_1$ (1,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (1,7), $x_1 \rightarrow x_3$ (0,1), $x_2 \rightarrow v$ (1,6), $x_3 \rightarrow v$ (1,2), and $x_4 \rightarrow x_3$ (1,4). Diagram 3 (Flow F_2): The flow values have updated again: $u \rightarrow x_1$ (3,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (3,7), $x_1 \rightarrow x_3$ (0,1), $x_2 \rightarrow v$ (3,6), $x_3 \rightarrow v$ (1,2), and $x_4 \rightarrow x_3$ (1,4).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0102	146	—	—	—	The top diagram shows a flow network with source u and sink v. Intermediate nodes are x_1, x_2, x_3, x_4. The flow F_3 is indicated. The bottom diagram shows the same network with the maximum flow F_4.
voloshin-hypergraph_ DIA_0103	151	—	—	—	A hypergraph diagram with five nodes and three hyperedges represented by ellipses.
voloshin-hypergraph_ DIA_0104	155	—	—	—	Two hypergraph diagrams labeled $\mathcal{H}_1$ and $\mathcal{H}_2$. $\mathcal{H}_1$ has five nodes and three hyperedges. $\mathcal{H}_2$ has six nodes and four hyperedges.

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																										
voloshin-hypergraph_ DIA_0105	157	—	—	—	The diagram illustrates the relationship between a hypergraph $\mathcal{H} = (X, \mathcal{D})$, its incidence matrix $I(\mathcal{H})$, and its bipartite graph $B(\mathcal{H})$. Hypergraph $\mathcal{H}$: A set of 6 vertices $\{1, 2, 3, 4, 5, 6\}$ and 5 hyperedges $\{D_1, D_2, D_3, D_4, D_5\}$. Vertex 1 is in D_1, vertex 2 is in D_2, vertex 3 is in D_3, vertex 4 is in D_4, and vertex 5 is in D_5. Vertex 6 is not in any hyperedge. Incidence Matrix $I(\mathcal{H})$: <table border="1"> <thead> <tr> <th></th><th>D_1</th><th>D_2</th><th>D_3</th><th>D_4</th><th>D_5</th></tr> </thead> <tbody> <tr> <th>1</th><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr> <tr> <th>2</th><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr> <tr> <th>3</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>4</th><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr> <tr> <th>5</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>6</th><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </tbody> </table> Bipartite Graph $B(\mathcal{H})$: A bipartite graph with vertices $\{1, 2, 3, 4, 5, 6\}$ on the left and $\{D_1, D_2, D_3, D_4, D_5\}$ on the right. Edges connect vertices to the hyperedges they belong to: $(1, D_1), (2, D_2), (3, D_3), (4, D_4), (5, D_5)$. Hypergraph $\mathcal{H}^*$: A set of 6 vertices $\{1, 2, 3, 4, 5, 6\}$ and 5 hyperedges $\{d_1, d_2, d_3, d_4, d_5\}$. Vertex 1 is in d_1, vertex 2 is in d_2, vertex 3 is in d_3, vertex 4 is in d_4, and vertex 5 is in d_5. Vertex 6 is not in any hyperedge.		D_1	D_2	D_3	D_4	D_5	1	1	1	1	0	0	2	0	1	1	1	0	3	0	0	0	1	1	4	0	0	1	0	1	5	0	0	0	1	1	6	0	0	0	0	0
	D_1	D_2	D_3	D_4	D_5																																										
1	1	1	1	0	0																																										
2	0	1	1	1	0																																										
3	0	0	0	1	1																																										
4	0	0	1	0	1																																										
5	0	0	0	1	1																																										
6	0	0	0	0	0																																										
voloshin-hypergraph_ DIA_0106	158	—	—	—	The diagram illustrates the relationship between a graph G and its dual graph G^*. Graph G: A graph with 5 vertices $\{1, 2, 3, 4, 5\}$ and 5 edges $\{a, b, c, d, e\}$. Vertex 1 is connected to 2 by edge a, vertex 2 is connected to 3 by edge b, vertex 3 is connected to 4 by edge c, vertex 4 is connected to 5 by edge d, and vertex 5 is connected to 1 by edge e. Dual Graph G^*: A graph with 5 vertices $\{a, b, c, d, e\}$ and 5 edges $\{1, 2, 3, 4, 5\}$. Vertex a is connected to b by edge 1, vertex b is connected to c by edge 2, vertex c is connected to d by edge 3, vertex d is connected to e by edge 4, and vertex e is connected to a by edge 5.																																										

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0107	161	—	—	—	
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voloshin-hypergraph_ DIA_0109	163	—	—	—	
voloshin-hypergraph_ DIA_0110	164	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0112	165	—	—	—	
voloshin-hypergraph_ DIA_0113	166	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0116	172	—	—	—	

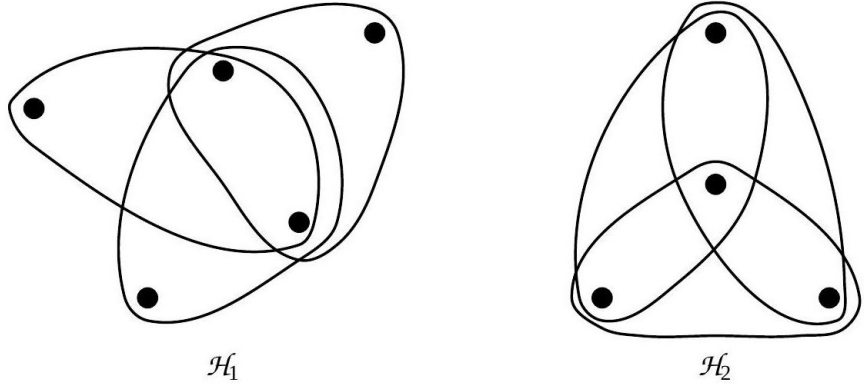
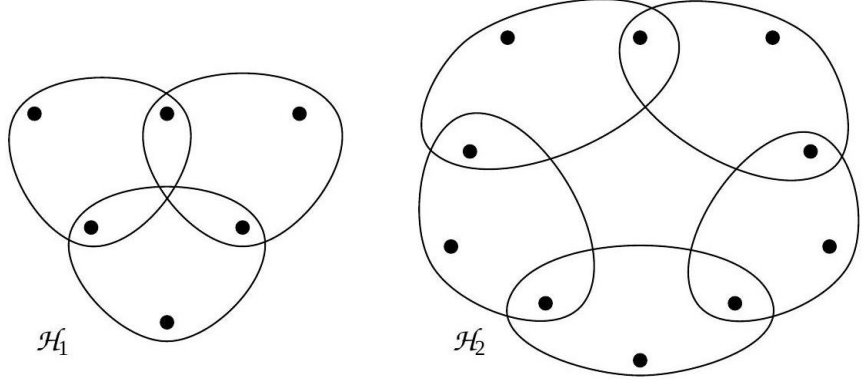
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voloshin-hypergraph_ DIA_0118	178	—	—	—	T C_4

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0120	180	—	—	—	$\mathcal{H}$ $\mathcal{H}_0$
voloshin-hypergraph_ DIA_0121	181	—	—	—	$\mathcal{H}_0^*$ $\mathcal{H}^*$
voloshin-hypergraph_ DIA_0122	181	—	—	—	$\mathcal{H}_0^*$ $\mathcal{H}^*$
voloshin-hypergraph_ DIA_0123	182	—	—	—	G $\mathcal{H}$ $\mathcal{H}^*$

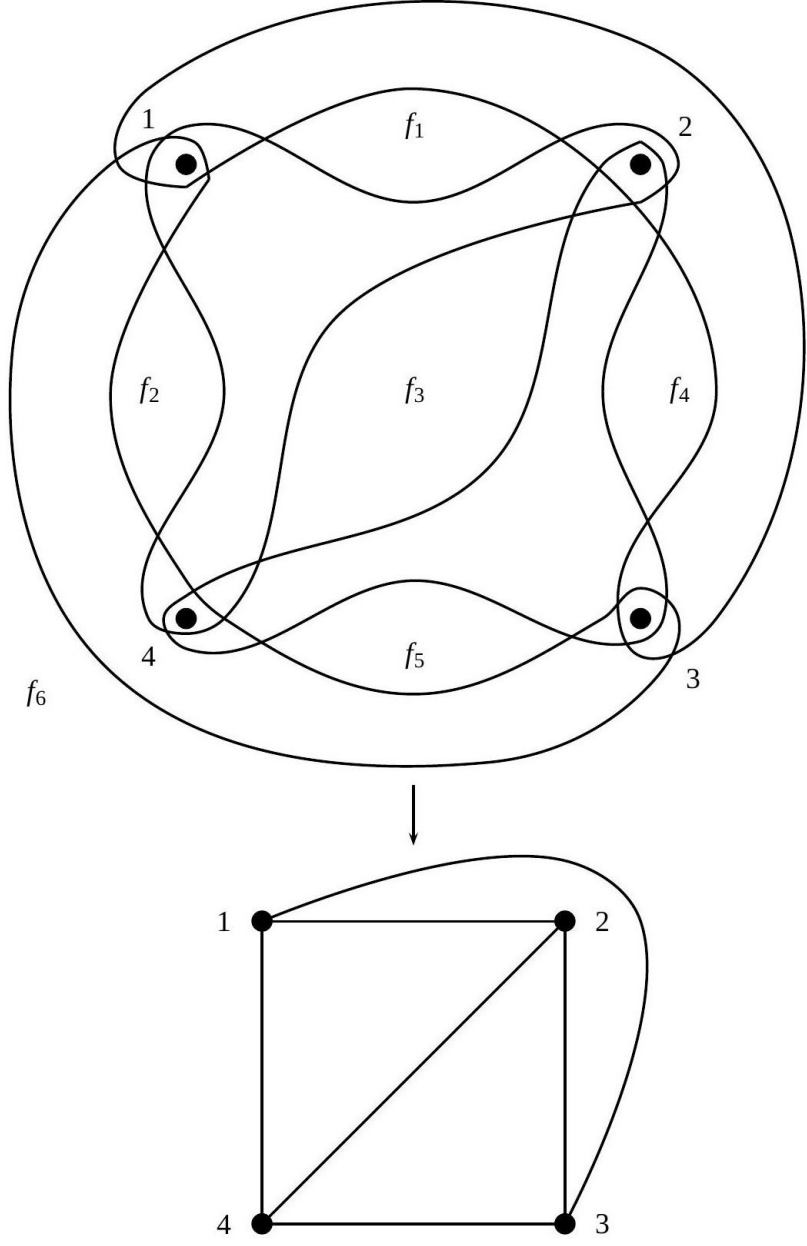
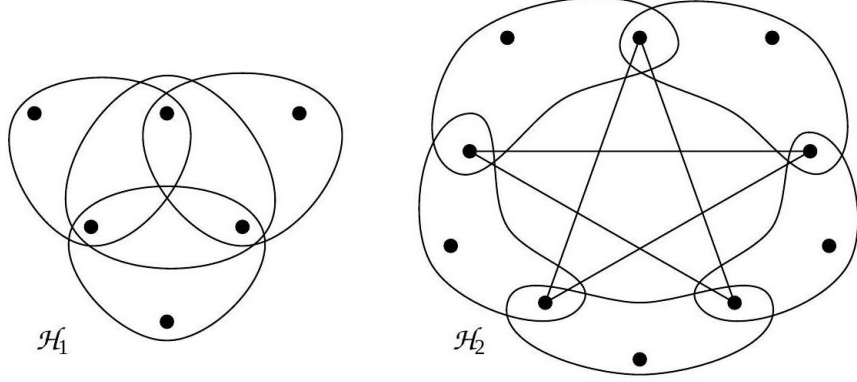
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0124	183	—	—	—	
voloshin-hypergraph_ DIA_0125	186	—	—	—	
voloshin-hypergraph_ DIA_0126	187	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0127	188	—	—	—	
voloshin-hypergraph_ DIA_0128	189	—	—	—	
voloshin-hypergraph_ DIA_0129	191	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0130	194	—	—	—	Diagram illustrating a hypergraph $\mathcal{H}$ and its relationship to a tree $T \subseteq [\mathcal{H}]_2$. The top part shows $\mathcal{H}$ with a vertex x and an arrow pointing to $T \subseteq [\mathcal{H}]_2$. The bottom part shows $\mathcal{H} - x$ and an arrow pointing to $T - x$.
voloshin-hypergraph_ DIA_0131	196	—	—	—	Diagram illustrating two hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$.
voloshin-hypergraph_ DIA_0132	198	—	—	—	Diagram illustrating a hypergraph structure with multiple vertices and hyperedges.
voloshin-hypergraph_ DIA_0133	199	—	—	—	Diagram illustrating two hypergraph structures, one above the other.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0134	200	—	—	—	 Diagram $\mathcal{H}_1$ shows three overlapping ellipses with four points. Diagram $\mathcal{H}_2$ shows three overlapping ellipses with three points.
voloshin-hypergraph_ DIA_0135	202	—	—	—	 Diagram $\mathcal{H}_1$ shows three overlapping ellipses with five points. Diagram $\mathcal{H}_2$ shows three overlapping ellipses with eight points.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0136	203	—	—	—	

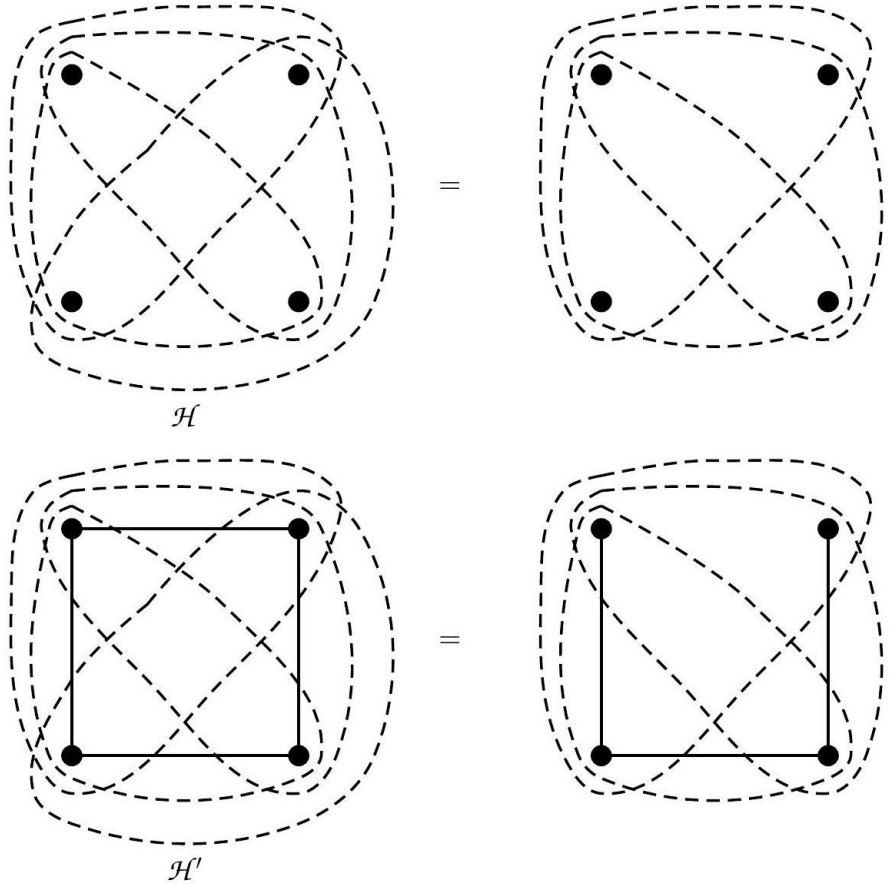
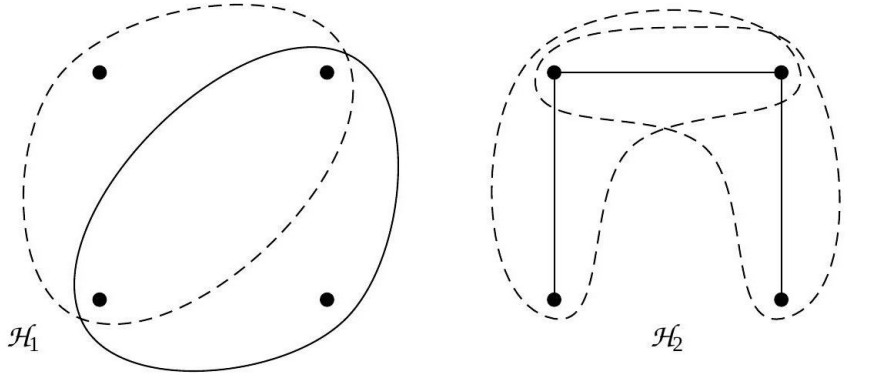
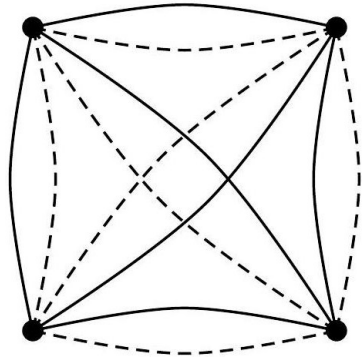
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0137	206	—	—	—	 The diagram illustrates a hypergraph with four vertices labeled 1, 2, 3, and 4. Six hyperedges, labeled f_1 through f_6, are shown as closed loops connecting these vertices. An arrow points down to a graph representation where the vertices are arranged in a square (1 at top-left, 2 at top-right, 3 at bottom-right, 4 at bottom-left). The edges of this graph correspond to the hyperedges: f_1 is the top edge (1-2), f_2 is the left edge (1-4), f_3 is the diagonal (1-3), f_4 is the right edge (2-3), f_5 is the bottom edge (3-4), and f_6 is a curved edge connecting 1 and 2.
voloshin-hypergraph_ DIA_0138	207	—	—	—	 Two hypergraphs are shown. $\mathcal{H}_1$ on the left has six vertices and three hyperedges represented by overlapping loops. $\mathcal{H}_2$ on the right has six vertices and a more complex set of hyperedges, including a central star-like structure and several outer loops.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0139	210	—	—	—	
voloshin-hypergraph_ DIA_0140	213	—	—	—	
voloshin-hypergraph_ DIA_0141	216	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0142	217	—	—	—	
voloshin-hypergraph_ DIA_0143	218	—	—	—	
voloshin-hypergraph_ DIA_0144	221	—	—	—	
voloshin-hypergraph_ DIA_0145	222	—	—	—	
voloshin-hypergraph_ DIA_0146	223	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0147	226	—	—	—	
voloshin-hypergraph_ DIA_0148	229	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0149	232	—	—	—	The diagram illustrates a sequence of operations on a hypergraph C (represented by a diamond shape with nodes 1, 2, 3, 4 and edges D_1, D_2, D_3, D_4): Top row: C is shown as a sum of two diamond shapes. The first diamond has edges D_1, D_2, D_3, D_4. The second diamond has edges D_1, D_2, D_3 and a dashed edge D_4. This is labeled "splitting 13". Second row: The first diamond is further split into two smaller diamonds (labeled "spl. 24"). The second diamond is transformed into a triangle with nodes 13, 4, 2 (labeled "C-clearing + contraction 13"). This is labeled "splitting 24". Third row: The first diamond is transformed into a triangle with nodes 13, 4, 2 (labeled "C-clearing + contraction 24"). The second triangle is transformed into a triangle with nodes 13, 4, 2 (labeled "contraction 24"). Bottom row: The final result is a sum of three components: a triangle with nodes 13, 4, 2, a triangle with nodes 13, 4, 2, and a triangle with nodes 13, 4, 2.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0150	234	—	—	—	
voloshin-hypergraph_ DIA_0151	235	—	—	—	
voloshin-hypergraph_ DIA_0152	237	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0153	238	—	—	—	
voloshin-hypergraph_ DIA_0154	239	—	—	—	
voloshin-hypergraph_ DIA_0155	242	—	—	—	
voloshin-hypergraph_ DIA_0156	242	—	—	—	$\mathcal{H}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0157	244	—	—	—	
voloshin-hypergraph_ DIA_0158	250	—	—	—	
voloshin-hypergraph_ DIA_0159	251	—	—	—	
voloshin-hypergraph_ DIA_0160	253	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0161	257	—	—	—	
voloshin-hypergraph_ DIA_0162	258	—	—	—	
voloshin-hypergraph_ DIA_0163	259	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0164	261	—	—	—	
voloshin-hypergraph_ DIA_0165	262	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0166	265	—	—	—	
voloshin-hypergraph_ DIA_0167	267	—	—	—	
voloshin-hypergraph_ DIA_0168	269	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0169	271	—	—	—	
voloshin-hypergraph_ DIA_0170	275	—	—	—	1st coloring: $ABBA$ 2nd coloring: $ABAB$ 3rd coloring: $AABB$ $\Rightarrow R(\mathcal{H}) = (0, 3, 0, 0), P(\mathcal{H}, \lambda) = 3\lambda(\lambda - 1)$ $\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0171	276	—	—	—	
voloshin-hypergraph_ DIA_0172	277	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0173	280	—	—	—	The diagram illustrates a mapping G from a graph to a hypergraph. The top graph has nodes Z_1, Z_2, Z_3 and edges $\{a, c\}, \{a, b\}, \{b, c\}$. The bottom hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ has nodes a, b, c and hyperedges Z_1, Z_2, Z_3. A double-headed arrow labeled G connects the two structures.
voloshin-hypergraph_ DIA_0174	281	—	—	—	The diagram illustrates a mapping G from a bipartite graph to a hypergraph. The top graph has nodes x_1, x_2, x_3, x_4 (jobs) and y_1, y_2, y_3, y_4, y_5 (resources) with various edges. The bottom hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ has nodes x_1, x_2, x_3, x_4 and hyperedges Z_1, Z_2, Z_3. A double-headed arrow labeled G connects the two structures.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0175	284	—	—	—	Induction basis Induction hypothesis Induction step 1. P_1 2. $P_k \rightarrow P_{k+1}$ $\longrightarrow P_n$ is true for all $n = 1, 2, \dots$
voloshin-hypergraph_ DIA_0176	284	—	—	—	Induction basis Induction hypothesis Induction step 1. P_1 2. $P_1, P_2, \dots, P_k \rightarrow P_{k+1}$ $\longrightarrow P_n$ is true for all $n = 1, 2, \dots$